

Debye screening and membrane charging

Why a screening time is not a charging time

Electrolytes screen equilibrium charge over a short distance, yet membrane voltages can evolve over much longer distances and times. The distinction becomes transparent if one first separates charge and salt in the Poisson–Nernst–Planck equations, and only then constructs a circuit limit. This note treats a passive, planar membrane; spatially localized channels require a different boundary-value problem.

1 Transport from electrochemical potentials

Take a dilute symmetric monovalent electrolyte at uniform temperature T . The number densities are c_+ and c_- , their common equilibrium value is c_0 , both species have diffusivity D , the electric potential is ϕ , and the solvent permittivity is ϵ . Let $e > 0$ be the elementary charge. Ignoring convection, reactions, and finite-ion-size effects, use

$$\mu_{\pm} = \mu_{\pm}^0 + k_B T \log(c_{\pm}/c_0) \pm e\phi, \quad (1)$$

$$\mathbf{J}_{\pm} = -\frac{Dc_{\pm}}{k_B T} \nabla \mu_{\pm} = -D \left(\nabla c_{\pm} \pm \frac{ec_{\pm}}{k_B T} \nabla \phi \right), \quad (2)$$

$$\partial_t c_{\pm} = -\nabla \cdot \mathbf{J}_{\pm}, \quad -\epsilon \nabla^2 \phi = e(c_+ - c_-). \quad (3)$$

The fluxes count ions per unit area per unit time. In particular, c_0 is not a concentration in moles per liter. These equations are a continuum model, not a molecular description of an ion channel. The classical charging problem is developed in Ref. [1].

Linearize about $c_{\pm} = c_0$, $\phi = 0$, so $c_{\pm} = c_0 + \delta c_{\pm}$ with small perturbations. Define the salt-density perturbation $s = \delta c_+ + \delta c_-$ and charge density $\rho = e(\delta c_+ - \delta c_-)$. Adding and subtracting the continuity equations gives

$$\partial_t s = D \nabla^2 s, \quad \partial_t \rho = D \left(\nabla^2 \rho - \frac{\rho}{\lambda_D^2} \right), \quad \lambda_D^2 = \frac{\epsilon k_B T}{2e^2 c_0}. \quad (4)$$

A bulk Fourier charge perturbation of wavevector magnitude q decays at rate $D(q^2 + \lambda_D^{-2})$. Salt has no analogous local relaxation term. Boundary conditions still determine which modes can occur: the local decay term does not allow net charge to disappear from a closed system. Integrating the original continuity equations always expresses total-charge change as boundary current.

2 Screening, conductivity, and capacitance

At equilibrium $\mathbf{J}_{\pm} = 0$. Linearized Boltzmann distributions imply $\rho = -2e^2 c_0 \phi / (k_B T)$. In an electrolyte occupying $z > 0$, with $\phi(0) = \phi_s$ and $\phi(\infty) = 0$, Poisson's equation becomes

$$\phi'' = \phi / \lambda_D^2, \quad \phi(z) = \phi_s e^{-z/\lambda_D}, \quad \int_0^{\infty} \rho dz = -\frac{\epsilon \phi_s}{\lambda_D}. \quad (5)$$

The diffuse charge opposes the charge on the bounding surface. The positive small-signal capacitance per area is therefore $C_D = \epsilon / \lambda_D$.

For uniform bulk concentrations, the electric current density is $e(\mathbf{J}_+ - \mathbf{J}_-) = -\sigma_b \nabla \phi$, where

$$\sigma_b = \frac{2e^2 D c_0}{k_B T}, \quad \tau_D = \frac{\lambda_D^2}{D} = \frac{\epsilon}{\sigma_b}. \quad (6)$$

Here σ_b is electrical conductivity, not a surface charge density. A nearly electroneutral bulk can conduct current: small charge density does not imply a vanishing electric field. A uniform field, for example, has zero divergence while driving ionic current.

Now insert a flat, impermeable dielectric membrane of thickness d and permittivity ϵ_m . Its capacitance per area is $C_m = \epsilon_m/d$. In a symmetric, weakly perturbed system with no fixed interfacial charge, the membrane and two diffuse layers form three capacitors in series:

$$\frac{1}{C_{\text{eff}}} = \frac{1}{C_m} + \frac{2}{C_D}. \quad (7)$$

All three capacitances in Eq. (7) are per area, with units F/m². For charge Q over area A , the total interfacial voltage is $V_{\text{int}} = Q/(AC_{\text{eff}})$, while the voltage inside the dielectric is $V_m = Q/(AC_m)$. The two voltages differ unless the membrane dominates the series capacitance. Treating a membrane as a surface with a prescribed capacitance is a reduced description of its finite-thickness electrostatics [2].

3 A circuit limit with stated boundaries

Place an ohmic electrolyte slab of thickness L on each side, both of area A . Ideal distant reservoirs impose the total voltage; neglect additional electrode impedances. Assume thin diffuse layers, a laterally uniform perturbation, negligible salt depletion, and diffuse layers that equilibrate locally faster than the overall charging. The bulk resistance is $R_b = 2L/(\sigma_b A)$, while the total capacitance is AC_{eff} . Charge conservation under a voltage step V_{app} gives

$$C_{\text{eff}} \frac{dV_{\text{int}}}{dt} = \frac{\sigma_b}{2L} (V_{\text{app}} - V_{\text{int}}), \quad (8)$$

$$\tau_{\text{RC}} = \frac{2LC_{\text{eff}}}{\sigma_b}, \quad V_{\text{int}}(t) = V_{\text{app}} \left(1 - e^{-t/\tau_{\text{RC}}}\right), \quad (9)$$

$$V_m(t) = \frac{C_{\text{eff}}}{C_m} V_{\text{int}}(t). \quad (10)$$

These expressions assume initially uncharged interfaces. At steady state the blocking membrane carries no ionic conduction current, so the bulk ohmic voltage drop vanishes. The stored interfacial charge remains.

Two limits expose the difference between screening and charging:

- If $C_m \ll C_D$, then $C_{\text{eff}} \simeq C_m$ and $\tau_{\text{RC}} \simeq 2LC_m/\sigma_b$.
- If $C_m \gg C_D$, then $C_{\text{eff}} \simeq C_D/2$ and $\tau_{\text{RC}} \simeq L\lambda_D/D$. Here L is one slab thickness; the full bulk path has length $2L$.

Neutral salt redistribution across a slab instead occurs on a scale $\tau_L = L^2/D$. Geometry-dependent prefactors belong to a specified boundary-value problem, not to this scaling estimate. Ref. [1] analyzes how strong forcing and salt adsorption complicate the simple circuit picture.

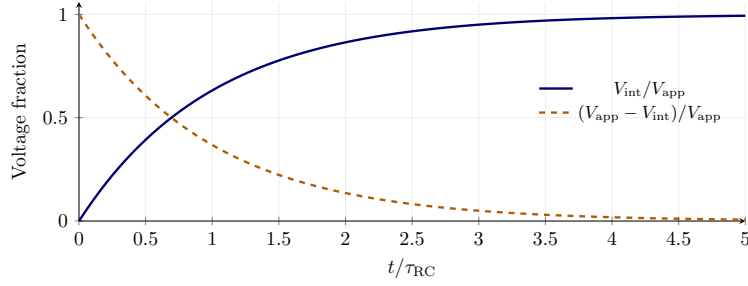


Figure 1: Uniform small-signal charging: interfacial voltage rises as the bulk voltage drop falls. Time is scaled by τ_{RC} . Analytic curves; no channel gating is included.

4 An order-of-magnitude calculation

Choose illustrative parameters $T = 298$ K, a salt concentration of 0.1 mol/L corresponding to $c_0 = (100 \text{ mol/m}^3)N_A$, $D = 10^{-9} \text{ m}^2/\text{s}$, $\epsilon = 78.5\epsilon_0$, $\epsilon_m = 2\epsilon_0$, $d = 5 \text{ nm}$, and $L = 10 \mu\text{m}$. Here N_A is Avogadro's constant and ϵ_0 the vacuum permittivity. Equations (6)–(9) give, to two significant figures,

λ_D	0.96 nm	σ_b	0.75 S/m
C_m	$3.5 \times 10^{-3} \text{ F/m}^2$	C_D	0.72 F/m ²
τ_D	0.92 ns	τ_{RC}	93 ns
τ_L	0.10 s	C_{eff}/C_m	0.99

This is a calculation for an idealized blocking membrane, not a prediction of a neuron's membrane time constant. A conducting membrane adds a parallel leak pathway and additional physics. The broad separation here also shows why replacing every electrical time by the Debye time is misleading.

5 Where this reduction stops

The one-dimensional circuit assumes uniform current over a fixed area. A localized channel creates lateral gradients and an access problem, so there is no reason to assign its dynamics the resistance $2L/(\sigma_b A)$ with an arbitrary area. Spatial membrane-interface models address that situation directly [3]. Large diffuse-layer potentials, unequal diffusivities, fixed surface charge, ion selectivity, and moving boundaries likewise require returning to transport and electrostatics with the appropriate boundary conditions. The circuit is valuable because its assumptions can be checked; it is not a replacement for them.

References

- [1] M. Z. Bazant, K. Thornton, and A. Ajdari, “Diffuse-charge dynamics in electrochemical systems,” *Phys. Rev. E* **70**, 021506 (2004). doi:10.1103/PhysRevE.70.021506.
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- [3] H. Row, J. B. Fernandes, K. K. Mandadapu, and K. Shekhar, “Spatiotemporal dynamics of ionic reorganization near biological membrane interfaces.” [arXiv:2407.11947](https://arxiv.org/abs/2407.11947) (2024).